NETWORKING

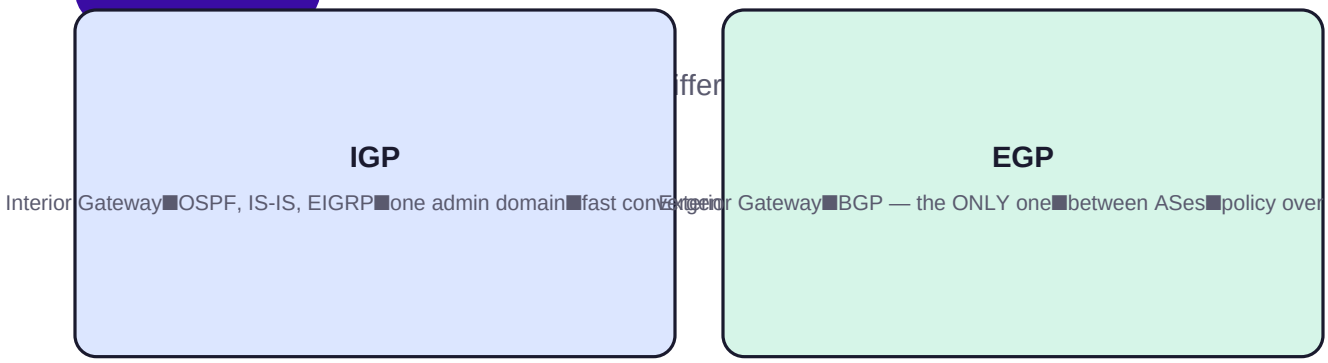
BGP & OSPF Visual Guide

How the internet routes —
link-state inside, path-vector between networks.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

IGP vs EGP: Two Different Jobs

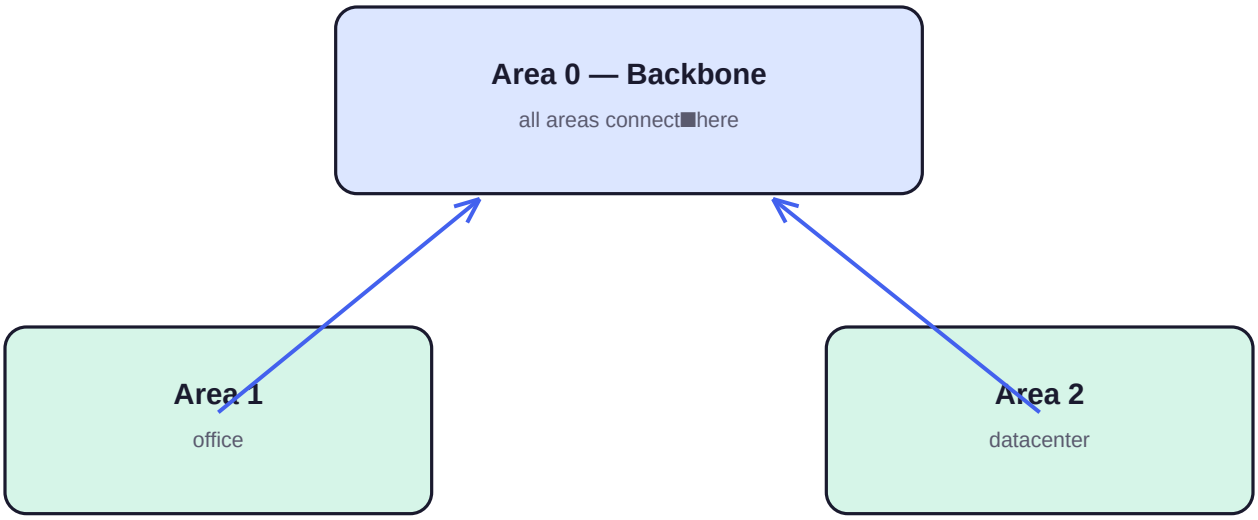


- IGP: you control every router — optimize for speed and shortest path.
- EGP: you control nothing beyond your border — optimize for POLICY (who you peer with, what you announce).
- An Autonomous System (AS) is a network under one admin — the internet is ~75k ASes stitched by BGP.

STEP 2/6

OSPF: Link-State & Areas

Every router learns the whole map, then runs Dijkstra.

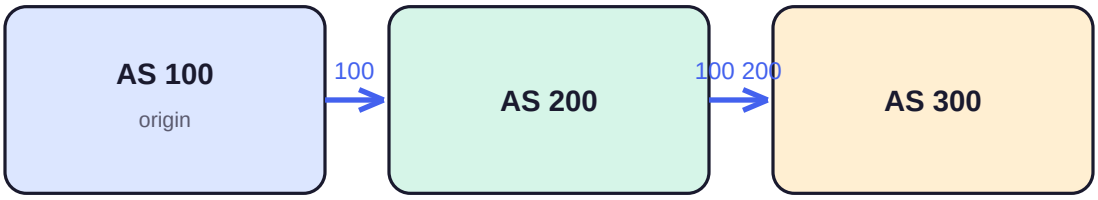


- Link-state: routers flood link costs; each builds an identical topology map and computes shortest paths locally.
- Areas contain the flooding — Area 0 (backbone) ties them together; ABRs summarize between areas.
- Converges in seconds; metric is cost (bandwidth-based). Hello packets maintain neighbor adjacencies.

STEP 3/6

BGP: Path Vector

BGP doesn't compute paths — it ANNOUNCES them, AS by AS.

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AS 300 learns:  'prefix P reachable via AS-path [300←200←100]'
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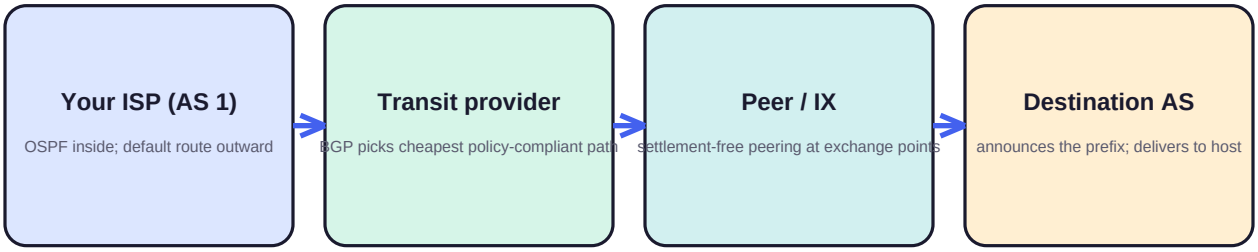
- AS-path = the list of ASes an announcement traversed; loop prevention is free (drop paths containing your own AS).
- Shortest AS-path usually wins — but policy (local-pref, MED) can override it.
- BGP runs over TCP/179 between border routers; full table is ~1M prefixes.

STEP 4/6

How a Packet Crosses the Internet

Hop by AS, decision by decision.



- Each AS makes its own independent routing decision — there is no central internet routing brain.
- Hot-potato routing: hand the packet to the next AS as quickly as possible.
- Traceroute shows this path router-by-router (when ICMP isn't filtered).

STEP 5/6

BGP Route Selection

Longest match first, then policy, then shortest path.

1. Longest prefix match

most specific route always wins

2. Highest local-pref

your policy: prefer this exit

3. Shortest AS-path

fewest AS hops

4. Lowest MED / IGP cost

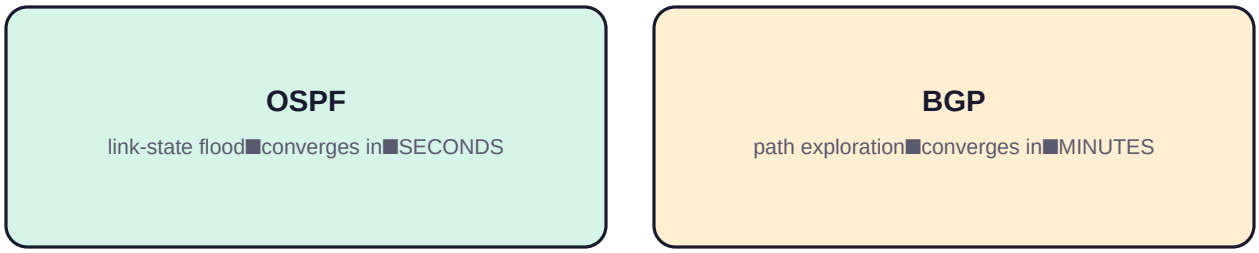
tie-breakers toward the exit

BGP hijack: announce someone else's prefix with a better path — traffic diverts to the attacker. RPKI/ROA validation is the defense.

STEP 6/6

Convergence: When Things Break

Routing protocols heal — at different speeds.



- Link failure inside your network: OSPF floods the change, all routers recompute — done in seconds.
- BGP failure: withdrawals propagate AS by AS; routers explore alternate paths — can take minutes globally.
- Design for it: redundant links, BFD for fast failure detection, anycast for instant failover.

STEP ✓/✓

Key Takeaways

BGP & OSPF — Remember

- IGP (OSPF) routes inside one admin domain; EGP (BGP) routes between Autonomous Systems.
- OSPF: link-state — every router maps the topology and runs Dijkstra; areas contain flooding.
- BGP: path-vector — AS-path announcements; loop-free by construction, policy-driven.
- Route selection: longest match → local-pref → shortest AS-path → tie-breakers.
- OSPF converges in seconds; BGP failover can take minutes — design redundancy accordingly.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev